CTIP: Towards Accurate Tabular-to-Image Generation for Tire Footprint Generation

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Introduction

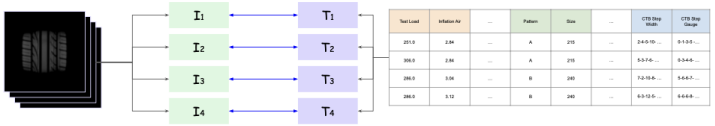
- Despite recent advancements in image generation models, their practical application remains limited, particularly in the manufacturing industry.
- While there have been studies under various conditions, such as text-to-image and image-to-image translation, research specifically focused on tabular-to-image generation using tabular data is still scarce.
- However, tabular-to-image studies, where tabular data is used as a condition, hold significant potential in manufacturing, especially in areas such as new product development and product testing.
- To the best of our knowledge, this is the first tabular-to-image generation study that uses tabular data as a condition to generate precisely targeted images.

Introduction

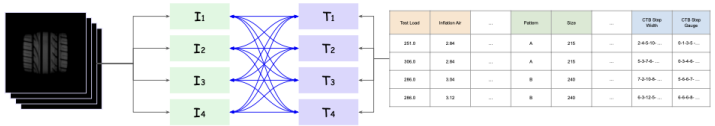
- Image generation models that simply embed tabular data as conditions face limitations in learning how changes in specific tabular features affect the image since they only train on one-to-one pairs between tabular data and images.

- This problem requires pre-training that integrates visual information to understand the impact of feature variations in tabular data on images. So, we introduce a novel approach, **C**ontrastive **T**abular-**I**mage **P**re-training (**CTIP**), inspired by the successes of CLIP.

- This pre-training method makes the tabular data more sensitive to capturing changes in the tabular feature, which is critical in manufacturing, where products are often tested under various conditions, and it is essential to produce desired results accurately.



Training an image generation model: Single data pair.

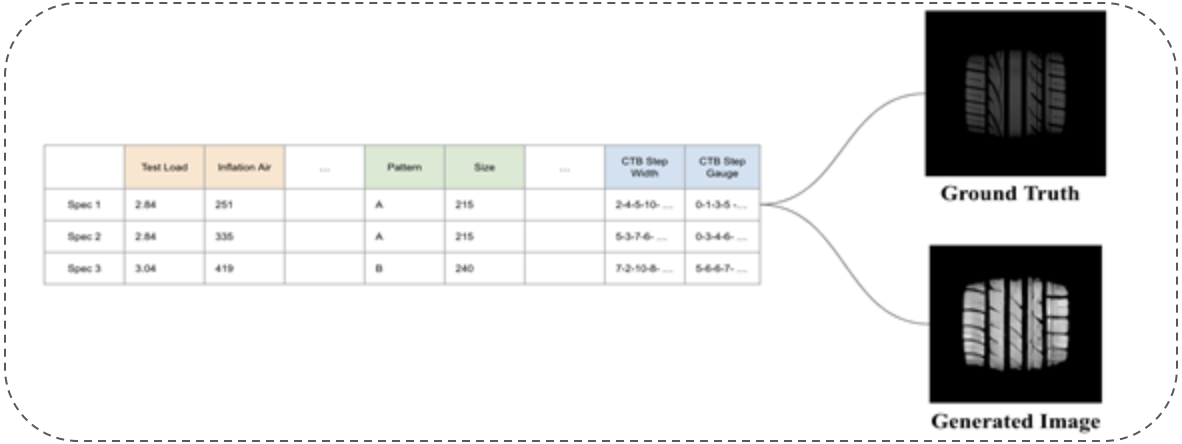


Training CTIP: Multiple data pair.

Introduction

- Unlike traditional image generation studies that aim to generate realistic images, we evaluate image generation performance by comparing it to ground truth (GT) images.

- Directly targeting and predicting tire performance metrics using tabular data does not explain the cause of the resulting values. For example, from the tabular data, users cannot see the resulting image, which makes it challenging to improve the problem because the current cause is unknown.



Contribution

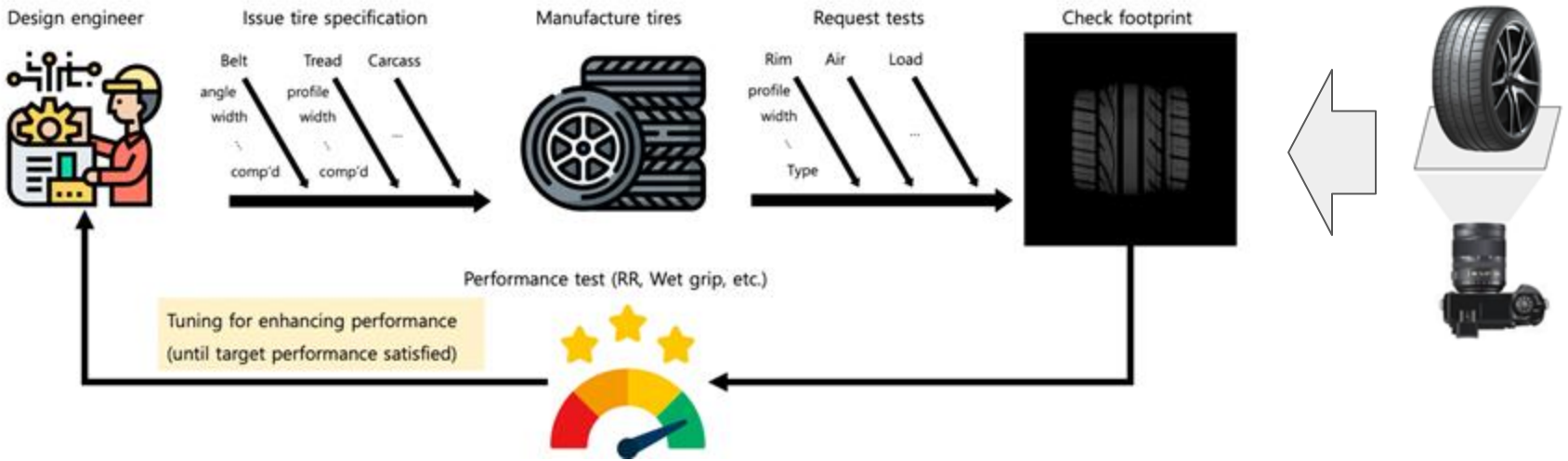
- We introduce a novel method, called CTIP (Contrastive Tabular-Image Pre-training), which applies the CLIP framework to pre-training multiple pairs of tabular data and images, improving a model's ability to produce accurate images from tabular data and presenting an example of its application to real-world manufacturing data.

- We show that CTIP effectively handles Feature Few-shot and Feature Zero-shot scenarios, generating high-quality and accurate images even when specific features are sparse or unseen, addressing common challenges in new product development.

- We demonstrate that CTIP generates high-quality embeddings that are more similar to the actual image information compared to simply embedding tabular data. Additionally, CTIP captures changes in specific features more sensitively and accurately.

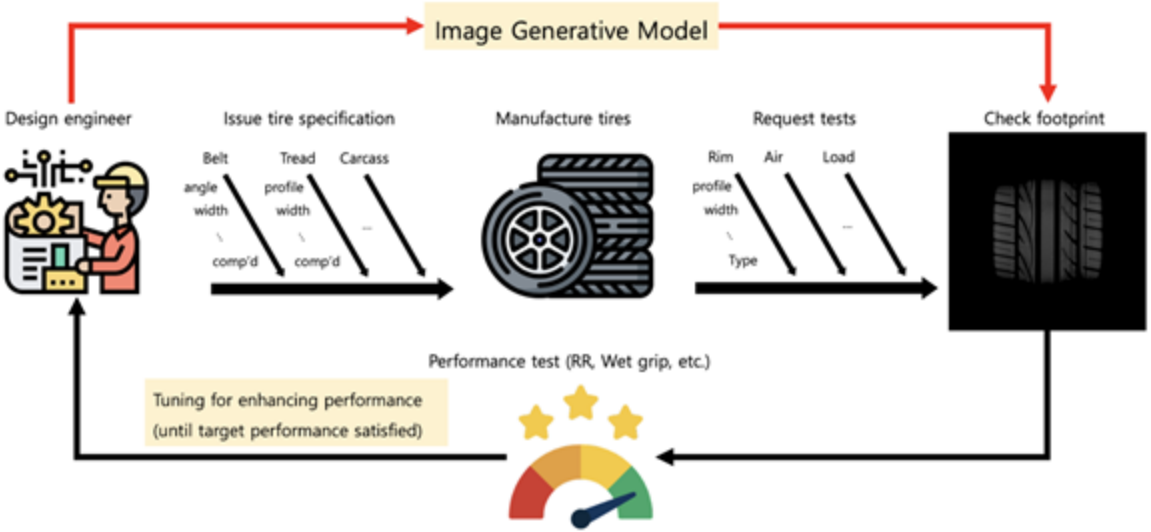
Problem Definition

- When tire developers develop a new tire, they set tire specifications and evaluate the performance of the tire based on tire footprint images taken under various test conditions.
- In general, it takes at least two months to obtain a tire footprint image, and tens of thousands of prototype tires are produced in a year.



Problem Definition

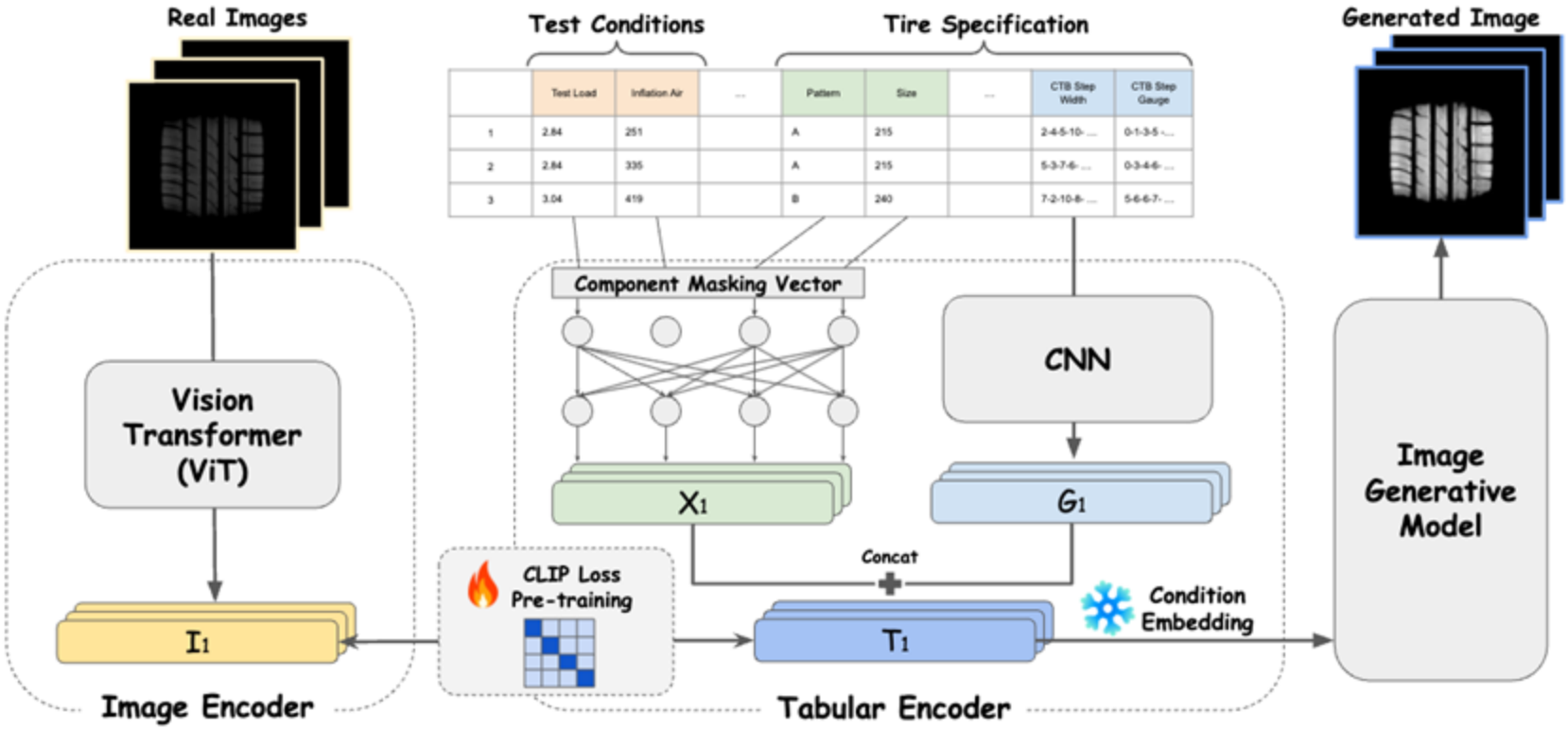
- We define the problem of this research as Tabular-to-Image, which predicts and generates a tire footprint image using tabular tire specifications and test conditions as input.
- If we can generate an accurate tire footprint image, we can improve the time efficiency of tire development and reduce the production of tires produced as prototypes.



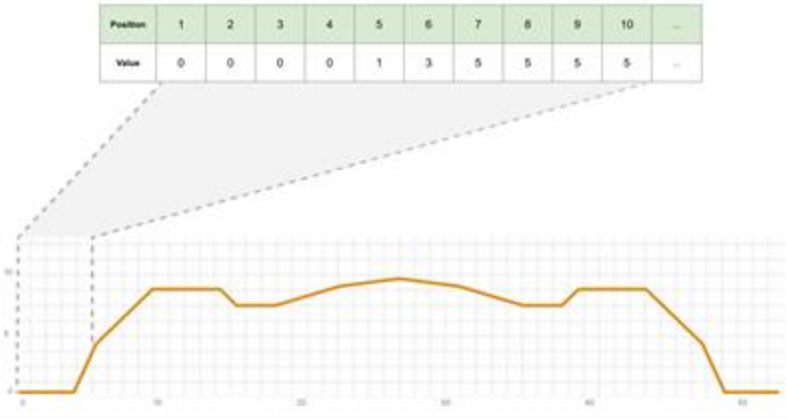
Contrastive Tabular-Image Pre-Training(CTIP)

- In this study, we propose Contrastive Tabular-Image Pre-Training (CTIP) to pre-train tabular data and images by applying the CLIP method, which was previously pre-trained using text and images.
- CTIP consists of an image encoder and a tabular encoder, and the image encoder and tabular encoder are trained simultaneously.
- Afterward, we freeze the trained Tabular Encoder and use the embedding generated by the Tabular Encoder as a condition for the Image Generation Model.
- We utilize two image generation models, DF-GAN and Latent Diffusion Model.

CTIP Main Figure



Experiments – Dataset



Dataset

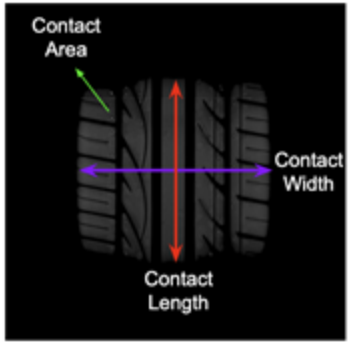
- Number of data: 64,756 images & tabular data pairs
- Image resolution: 1200x1200 -> 256x256
- Image scale: Gray scale
- Tabular data
  - Spec feature: 113
  - Test condition: 3

Feature Name	Tire Spec or Test Condition	Type	Example
Pattern	Tire Specification	Category	H826
Tire Size	Tire Specification	Numeric	250
...	...	...	...
CTB_Step_Width	Tire Specification	Graph	5.0-3.0-7.0-6.0-...
CTB_Step_Gauge	Tire Specification	Graph	0.0-0.2-0.2-0.2-...
...	...	...	...
Inflation_air	Test Condition	Numeric	3.4
Test_load	Test Condition	Numeric	256

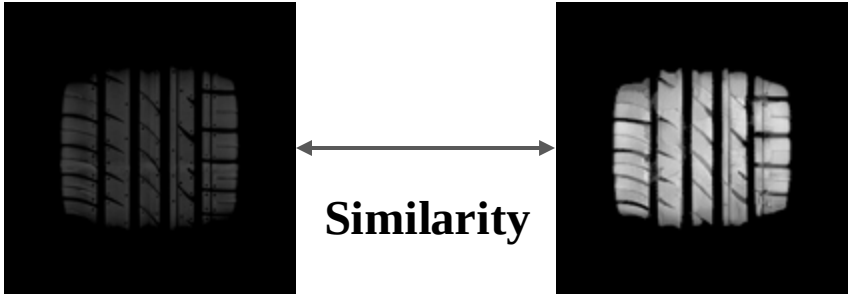
Experiments – Metric

Evaluation Metric

- Tire Performance
  - Contact Length
  - Contact Width
  - Contact Area



- Image Similarity
  - LPIPS(Learned Perceptual Image Patch Similarity)
  - SSIM(Structural Similarity)
  - PSNR(Peak Signal-to-Noise Ratio)



Experiment – Result

- The results demonstrate superior performance when using tabular data with a tabular encoder instead of converting tabular data to text.

- CTIP showed significant performance improvements in both DF-GAN and LDM.

Table 1. Tire performance & Image similarity result

Data Type	Models	Performance			Similarity		
		CL↓	CW↓	CA↓	LPIPS↓	SSIM↑	PSNR↑
Text	DF-GAN w/o Pre-train	27.86%	16.28%	40.11%	0.3093	0.7022	13.374
	DF-GAN with CLIP	20.26%	11.79%	28.04%	0.2478	0.7392	15.102
	LDM w/o Pre-train	30.01%	17.62%	47.49%	0.3279	0.6767	12.952
	LDM with CLIP	13.24%	7.78%	19.03%	0.2019	0.7534	16.045
Tabular	DF-GAN w/o Pre-train	16.96%	7.50%	25.18%	0.1857	0.7638	16.357
	DF-GAN with TB	12.80%	6.82%	17.28%	0.1629	0.7689	16.645
	DF-GAN with CTIP	5.83%	3.86%	8.02%	0.1291	0.7777	17.104
	LDM w/o Pre-train	9.79%	5.52%	15.67%	0.1521	0.7674	16.668
	LDM with TB	8.12%	5.15%	12.08%	0.1406	0.7320	16.975
	LDM with CTIP	**3.29%**	**3.65%**	**5.55%**	**0.1154**	**0.7891**	**17.666**

Experiment – Result

- The scatter plot in Figure 3 further illustrates improved predicted values and performance metrics for DF-GAN and LDM.
- It is evident that the scatter plot in orange color, representing the results after applying CTIP, clusters more closely around the center line compared to the blue scatter plot without CTIP.
- Overall, CTIP significantly improved the quality of the images generated, and in all cases, the results were more favorable than processing tabular data with text using CLIP.

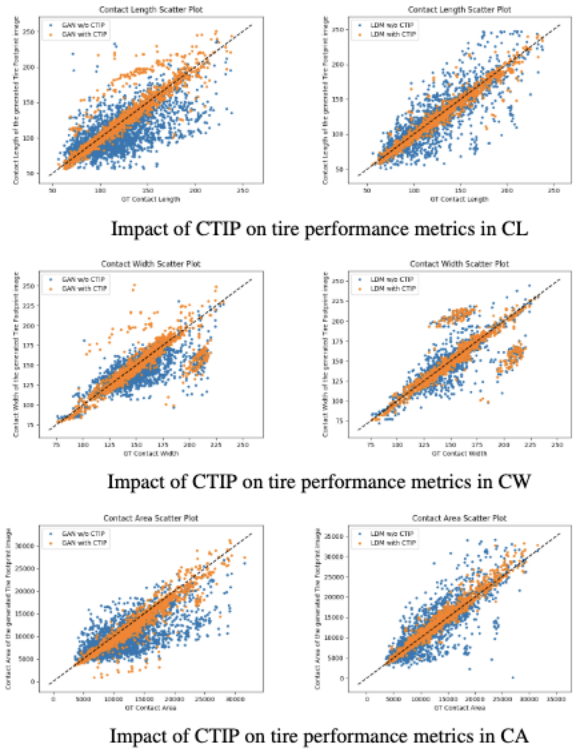
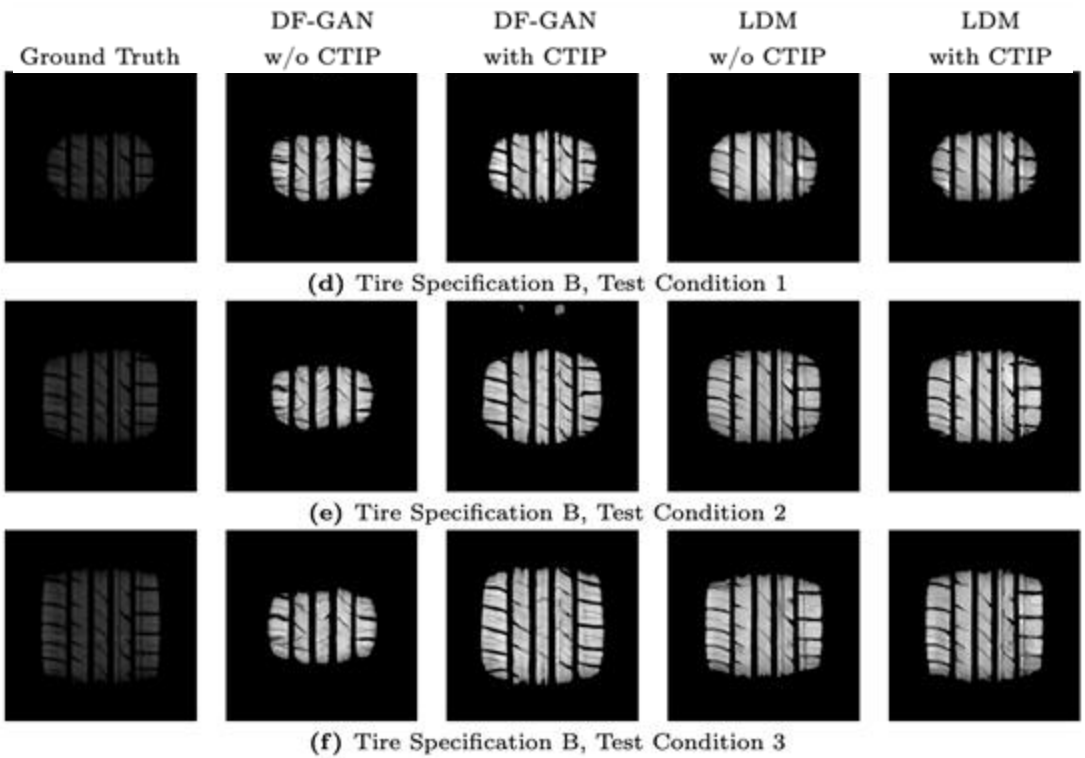


Figure 3. The plot for DF-GAN is shown on the left and the plot for LDM on the right. The vertical axis is the value of the generation image and the horizontal axis is the value of the ground truth. The blue color represents the values without CTIP and the orange color represents the values with CTIP.

Experiment – Result



(d) Tire Specification B, Test Condition 1

(e) Tire Specification B, Test Condition 2

(f) Tire Specification B, Test Condition 3

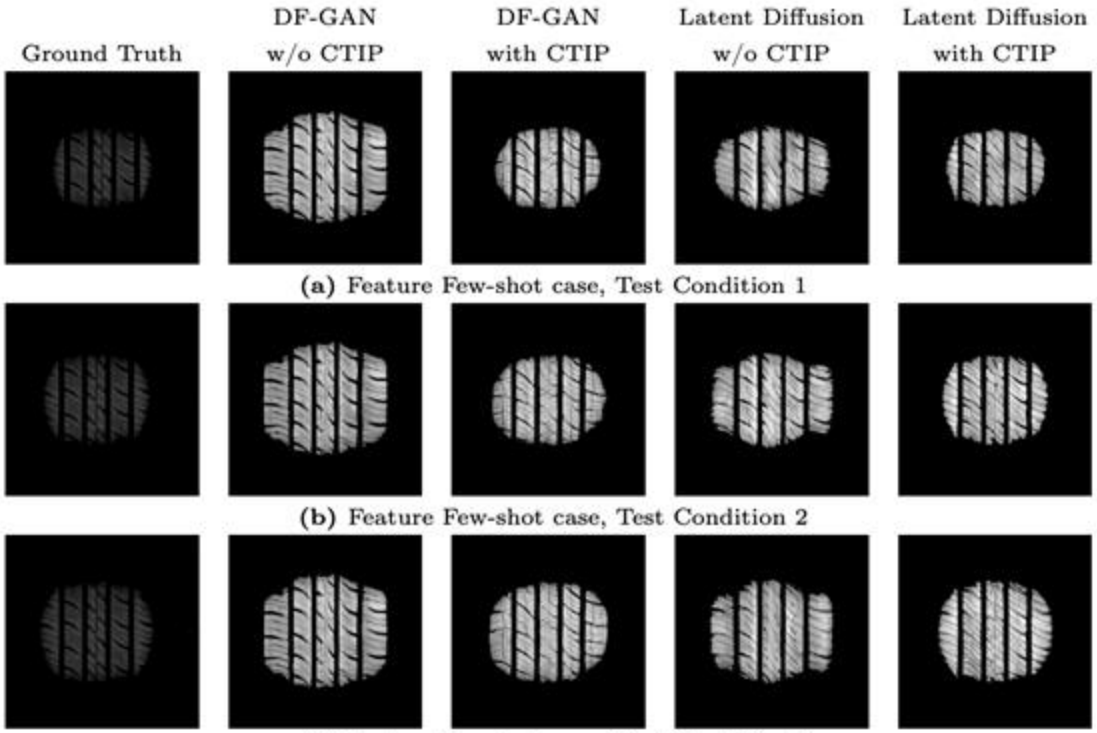
Analysis – Feature Few & Zero-shot Results

- We experimented with defining rarely used cases of Patterns as Feature Few & Zero-shot to see how well they robustly generate images.
  - 5 or less Patterns case -> Feature Few Shot
  - Unseen Pattern case -> Feature Zero Shot
- We can see that when the train set lacks certain patterns, the results without CTIP are not good in all metrics, but when with CTIP, it generates images robustly.

Table 2. Tire performance results for feature few-short and zero-shot

Data Type	Models	Few-shot			Zero-shot		
		CL↓	CW↓	CA↓	CL↓	CW↓	CA↓
Text	DF-GAN w/o Pre-train	28.64%	16.64%	39.82%	28.45%	16.48%	41.90%
	DF-GAN with CLIP	24.55%	15.90%	30.87%	27.40%	15.24%	37.03%
	LDM w/o Pre-train	32.88%	18.17%	50.46%	24.96%	18.22%	39.70%
	LDM with CLIP	17.86%	9.60%	24.69%	18.94%	13.49%	27.53%
Tabular	DF-GAN w/o Pre-train	21.46%	10.20%	30.96%	26.11%	7.79%	27.00%
	DF-GAN with TB	18.64%	8.75%	29.82%	28.45%	8.48%	24.12%
	DF-GAN with CTIP	4.30%	4.26%	7.44%	13.88%	8.39%	18.85%
	LDM w/o Pre-train	12.66%	6.38%	20.43%	19.47%	8.68%	30.99%
	LDM with TB	10.12%	6.08%	18.41%	16.19%	6.12%	24.11%
	LDM with CTIP	**3.73%**	**4.46%**	**7.18%**	**6.51%**	**5.69%**	**7.64%**

Analysis – Feature Few Results

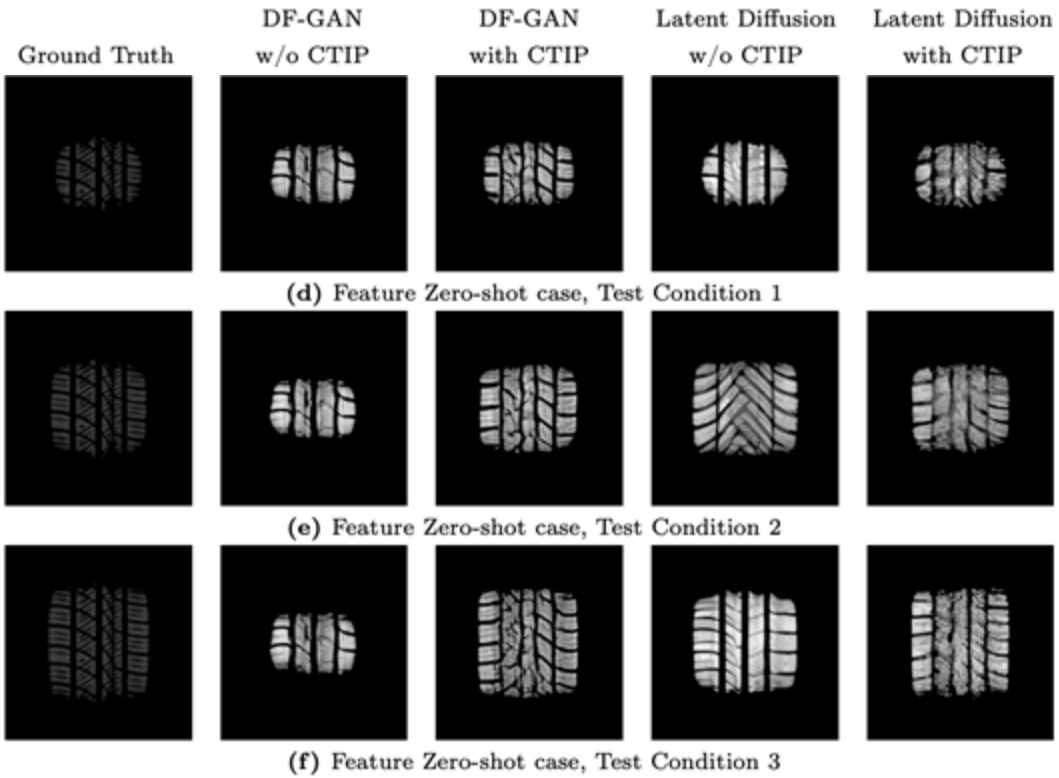


(a) Feature Few-shot case, Test Condition 1

(b) Feature Few-shot case, Test Condition 2

(c) Feature Few-shot case, Test Condition 3

Analysis – Zero-shot Results



(d) Feature Zero-shot case, Test Condition 1

(e) Feature Zero-shot case, Test Condition 2

(f) Feature Zero-shot case, Test Condition 3

Analysis – Impact of Test Condition

- As the Test Condition increases from 1 to 2 to 3, the Tire Footprint Image should also increase.
- However, without CTIP, it appears that the Test Condition is not captured accurately, and the three Test Conditions produce similarly sized Tire Footprint Images.
- With CTIP, on the other hand, the differences across Test Conditions are clearly reflected.

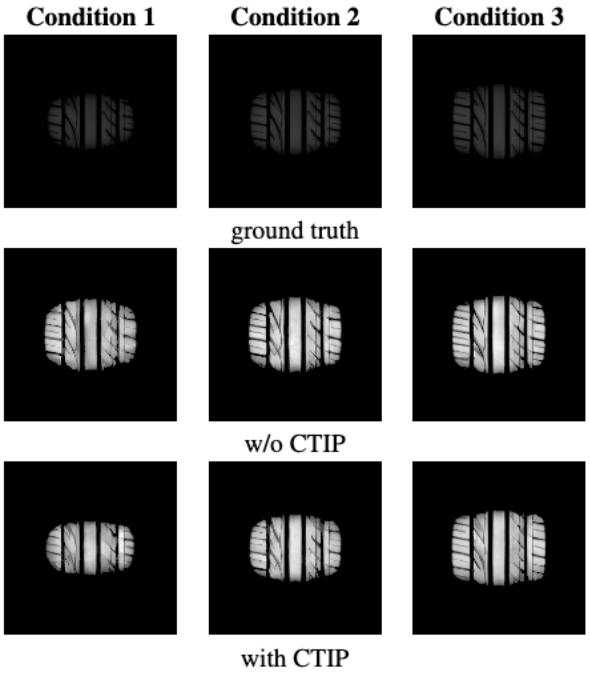


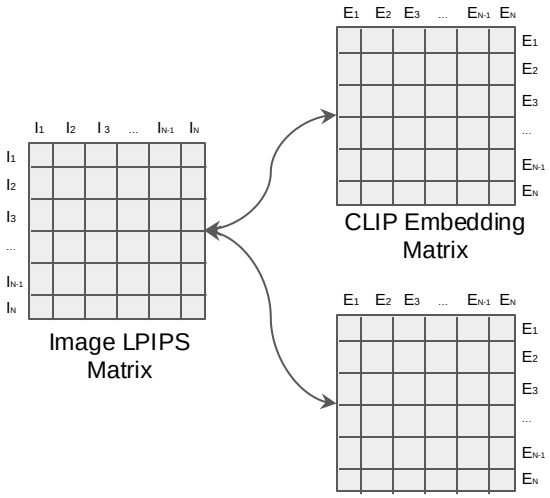
Figure 5. It shows the change of the tire footprint according to the Test Condition.

Analysis – Evaluating CTIP through Image Similarity

- To verify that CTIP produces Condition Embeddings that are similar to the actual Ground Truth image, we used the following methods
  - Measuring similarity using LPIPS: We generated a similarity matrix representing the similarity to the Ground Truth image.
  - Calculated Euclidean Distance Matrix: We calculated the Euclidean distance matrix between each Condition Embedding with and without CTIP.
  - Compare Cosine Similarity: We evaluated the effectiveness of CTIP by comparing the cosine similarity between each matrix.

- This shows that CTIP projects the condition embeddings similarly to the actual image information, which is key to generating high quality embeddings.

$$CosineSim(S_{\text{LPIPS}}, D_{\text{w\textbackslash o CTIP}}) : \underline{0.8832}$$
$$CosineSim(S_{\text{LPIPS}}, D_{\text{with CTIP}}) : \underline{0.9365}$$



$$S_{\text{LPIPS}} = \text{LPIPS}(I_{\text{gt}}, I_{\text{gt}}) \quad (1)$$
$$D_{\text{w\textbackslash o CTIP}} = \text{ED}(C_{\text{w\textbackslash o CTIP}}, C_{\text{w\textbackslash o CTIP}}) \quad (2)$$
$$D_{\text{with CTIP}} = \text{ED}(C_{\text{with CTIP}}, C_{\text{with CTIP}}) \quad (3)$$

, where I_{gt} represents the GT tire footprint images and $C_{\text{w\textbackslash o CTIP}}$, $C_{\text{with CTIP}}$ represents the condition embedding without and with CTIP.

Conclusion & Future work

- The limitation of this study is that we used only a single dataset to validate CTIP due to the lack of comparable open datasets in this specific context.

- The tabular data is not tabularized with information about the paired images but features about the tire and the test condition. Additionally, the features in the tabular data are mostly interdependent rather than independent.

- Future research should explore the potential of applying CTIP to tabular-to-image conversion tasks across a broader range of domains

- However, the novel tabular-to-image approach proposed with CTIP may inspire further discoveries in the industry and encourage the release of relevant datasets.

Thank you

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